



Jihyeon Son

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| PUBLICATIONS

- **Six-hour Prediction of Interplanetary Magnetic Field Bz Profiles for Strong Southward Cases by Deep Learning**
[Jihyeon Son](#), Yong-Jae Moon, Young-Sil Kwak, Kyung Sun Park, and Hyun-Jin Jeong
The Astrophysical Journal, 984(1), 67. (2025)
- **Three-day Forecasting of Solar Wind Speed Using SDO/AIA Extreme-ultraviolet Images by a Deep-learning Model**
[Jihyeon Son](#), Suk-Kyung Sung, Yong-Jae Moon, Harim Lee, and Hyun-Jin Jeong
The Astrophysical Journal Supplement Series, 267(2), 45. (2023)
- **72-Hour Time Series Forecasting of Hourly Relativistic Electron Fluxes at Geostationary Orbit by Deep Learning**
[Jihyeon Son](#), Yong-Jae Moon, and Seungheon Shin
Space Weather, 20(10), e2022SW003153. (2022)
- **Generation of He I 1083 nm Images from SDO AIA Images by Deep Learning**
[Jihyeon Son](#), Junghun Cha, Yong-Jae Moon, Harim Lee, Eunsu Park, Gyungin Shin, and Hyun-Jin Jeong
The Astrophysical Journal, 920(2), 101. (2021)
- **Comparison of Empirical and Deep Learning Models for Solar Wind Speed Prediction**
Seungwoo Ahn, [Jihyeon Son](#), Yong-Jae Moon, and Hyun-Jin Jeong
The Astrophysical Journal, 987(2), 179. (2025)
- **Solar cycle Dependence of NOAA Space Weather Scale Frequencies**
Daeil Kim, Yong-Jae Moon, Hyun-Jin Jeong, [Jihyeon Son](#)
Journal of Korean Astronomical Society, 59(1), 55-61. (2025)
- **3D Magnetic Free Energy and Flaring Activity Using 83 Major Solar Flares**
Khojiakbar Karimov, Harim Lee, Hyun-Jin Jeong, Yong-Jae Moon, Jihye Kang, [Jihyeon Son](#), Mingyu Jeon, Kanya Kusano

The Astrophysical Journal Letters, 965(1), L5. (2024)

• **Application of Deep Learning to Solar and Space Weather Data**

Yong-Jae Moon, Harim Lee, **Jihyeon Son**, Suk-Kyung Sung, Kangwoo Yi, Hyun-Jin Jeong, Eunsu Park, Eun-Young Ji, Il-Hyun Cho, Bendict Lawrance, Daye Lim, Gyungin Shin, Sujin Lee, Sumiaya Rahman and Taeyoung Kim
Proceedings of the International Astronomical Union, 18, S372 (2023)

EDUCATION

2020.03 – 2024.02.

Combined Master’s and Doctoral Course in Solar Physics

School of Space Research, Kyung Hee University, Republic of Korea

2015.03 – 2020.02.

Bachelor of Science in Space science

Department of Astronomy & Space Science, Kyung Hee University, Republic of Korea

EMPLOYMENT

2025.05. – present

Postdoctoral Researcher

Center for Heliophysics Research, Korea Astronomy & Space science Institute, Republic of Korea

2024.03. – 2025.04.

Postdoctoral Researcher

Astronomy & Space Science, College of Applied Science, Kyung Hee University, Republic of Korea

PATENTS

2023.07.

- Apparatus for predicting Solar Wind Speed using Deep Learning Model and Method thereof
Yong-Jae Moon and **Jihyeon Son**
10-2023-0087337, Republic of Korea Patent Application

AWARDS

2020.12.

- **Grand Prize**, Korean Space Weather Center: Artificial Intelligence (AI) competition for space weather forecasting

PARTICIPATED PROJECTS

2024.05. – present

- **Development of prediction models for solar wind parameters and geomagnetic activity using deep learning**
Role: Principal Investigator

2025.05. – present

- **Research on next-generation space weather variation and prediction**
Role: Development of a solar wind forecasting model

		2023.01. – 2023.12.
	• Study on the solar wind forecast and 3D ionospheric modelling improvement using deep learning	
	Role: Development of solar wind forecasting model	
		2023.01. – 2023.12.
	• Study on the forecast of solar winds and IGS 3D ionospheric modelling technique using deep learning	
	Role: Development of solar wind speed forecasting model	
		2021.12. – 2024.12.
	• Development of analysis and forecast models for space weather operations	
	Role: Development of space weather forecasting model	
		2020.05 – 2022.02
	• Study on the forecast of the occurrence, strength, and temporal evolution of solar flares using deep learning	
	Role: Development of flare forecasting model	
SCHOOL PROGRAM		2022.05.
	• Python in Heliophysics Summer School <i>Madrid, Spain</i>	
TECHNICAL REPORTS	• Python	
	• Analysis of solar image data and solar wind data	
	• Deep learning: Tensorflow keras & Pytorch	
	• Microsoft (Word, Excel, Powerpoint)	